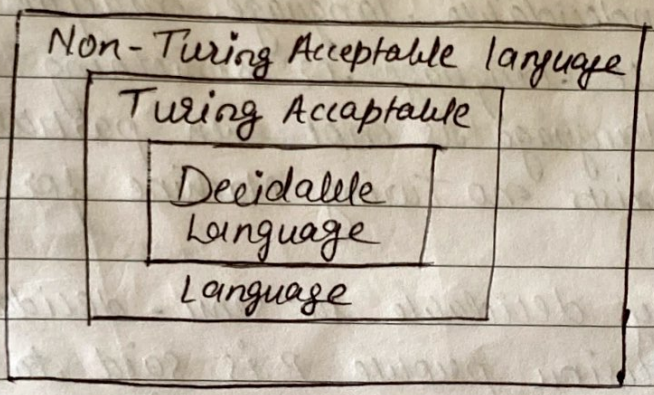
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Module 6 : Undecidability

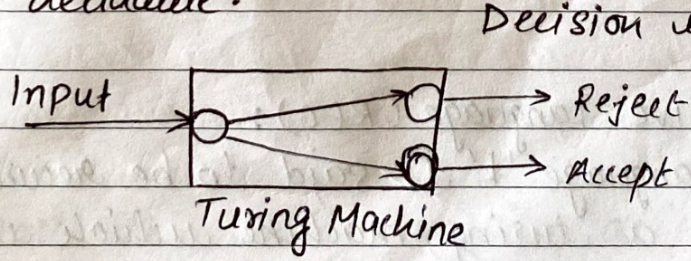
Decidability and Undecidability, Recursive & Recursively Enumerable Languages, Halting Problem, Rice's Theorem, Post Correspondence Problem.

⇛ Decidability and Undecidability :

A language is called decidable or recursive, if there is a Turing machine which accepts and halts on every input string w.
Every decidable language is Turing - Acceptable.



- **Decidable Language :**
  - A decidable problem P is said to be decidable, if the language L of all acceptable (yes) instances to P is decidable.



- For a decidable language, for each input string, the TM halts either at accept or reject state, as shown in figure above.

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• Example:
i. (Acceptance problem for DFA) Given a DFA does it accept a given word?
ii. (Emptiness problem for DFA) Given a DFA does it accept any word?
iii. (Equivalence problem for DFA) Given two DFAs, do they accept the same language?

- Undecidable Language:
• A decision problem P is said to be undecidable if the language L of all acceptable (yes) instances to P is not decidable or a language is undecidable if it is not decidable.
• An undecidable language maybe a partially decidable language or something else but not decidable.
• If a language is not even partially decidable, then there exists no Turing Machine for that language.

- Partially decidable or semi-decidable language:
• A decision proble P is said to be semi-decidable (i.e. have partial algorithm) if the language L of all yes instances to P is RE.
• A language L is partially decidable if 'L' is RE but not REC language.

Recursive Language (REC):
• A language 'L' is said to be recursive, if there exists a turing machine which will accept all the strings in 'L' and reject all the strings not in L.

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- The turing machine will halt every time and gives an answer (accept or reject) for each and every string input.
- A language 'L' is decidable if it is a recursive language.
- All decidable languages are recursive language & vice-versa.

- Recursively enumerable language (RE):
- A language 'L' is said to be a recursively enumerable language 'L' is said to be a recursively enumerable language if there exists a turing machine which will accept (and therefore halt) for all the input strings which are in 'L' but may or may not halt for input strings which are not in 'L'.
- By definition, all REC language are also RE languages but not all RE languages are REC languages.

Halting Problem:

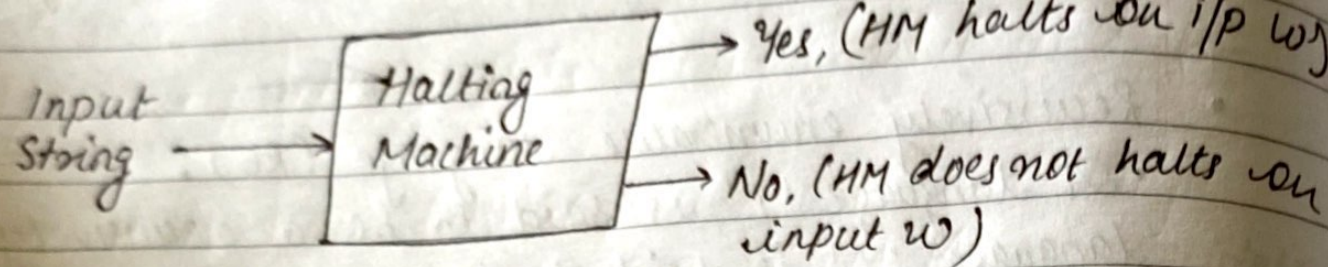
Input - A turing machine and an input string w.

Problem - Does the turing machine finish computing of the string 'w' in a finite number of steps? The answer must be either yes or no.

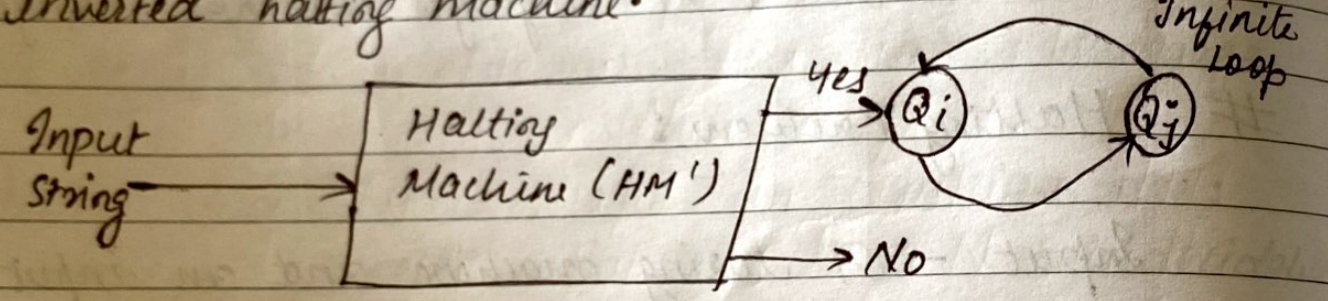
Proof -
- Initially assume that such turing machine exists to solve this problem and then, contradict it.
- Let's call this turing machine as 'Halting machine' that produces either 'yes' or 'no' in finite amount of time.

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- If the halting machine finishes in a finite amount of time, the output comes as 'yes', otherwise as 'no'.
- The following is the block diagram of a Halting machine:



- Now lets design an inverted halting machine (HM)' as -
  - If HM' returns Yes, then loop forever.
  - If HM' returns No, then halt.
- The following is the block diagram of an inverted halting machine:



- Further a machine $(HM)_2$ which i/p itself is constructed as follows -
  - If $(HM')_2$ halts on i/p, then loop forever.
  - Else, halt.
- Here, we have got a contradiction. Hence, the halting problem is undecidable.

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Rice's Theorem :

- Rice's Theorem states that any non-trivial semantic property of a language which is recognized by Turing Machine is undecidable.
- A property P, is the language of all Turing machines that satisfy that property.

Formal Definition :

- If P is a non-trivial property and the language holding the property L_P, is recognized by Turing machine M, the $L_P = \{\langle M \rangle \mid L(M) \in P\}$ is undecidable.

Description and Properties :

- Property of languages, P is simply a set of languages. If any language belongs to P ($L \in P$), it is said that L satisfies the property P.
- A property is called to be trivial if either it is not satisfied by any recursively enumerable languages, or if it is satisfied by all recursively enumerable languages.
- A non-trivial property is satisfied by some recursively enumerable languages and are not satisfied by others. In a non-trivial property where $L \in P$, both the following properties hold:
  - Property 1 : There exists TM, M1 & M2 that recognize the same language, i.e. either ($\langle M1 \rangle, \langle M2 \rangle \in L$) or ($\langle M1 \rangle, \langle M2 \rangle \notin L$)

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- Property 2- There exists TMs M_1 & M_2 where M_1 recognizes the language while M_2 does not i.e. $\langle M_1 \rangle \in L$ & $\langle M_2 \rangle \notin L$

Proof:

Suppose a property P is non-trivial and $\varphi \in P$.

- Since P is non-trivial, at least one language satisfies P, i.e. $L(M_0) \in P$, $\Rightarrow$ TM M_0.
- Let 'w' be an input in a particular instant and N is a TM which follows:
- One input x -
  - Run M on w.
  - If M does not accept (or doesn't halt) then do not accept x (or do not halt)
  - If M accepts w then run M_0 on x. If M_0 accepts x then accept x.
- A function that maps an instance $A_{TM} = \{\langle M, w \rangle \mid M \text{ accepts input } w\}$ to a N such that,
  - If M accepts 'w' and N accepts the same language as M_0, then $L(M) = L(M_0) \in P$.
  - If M does not accept w and N accepts φ, then $L(N) = \varphi \notin P$.

Since A_{TM} is undecidable and it can be reduced to L_P, L_P is also undecidable.

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Post correspondence Problem:

- The Post Correspondence Problem (PCP), introduced by Emil Post in 1946, is an undecidable decision problem.
- The PCP problem over an alphabet Σ is stated as follows:

Given the following two lists, M & N of non-empty strings over Σ -

$$M = (x_1, x_2, x_3, \cdots, x_n)$$
$$N = (y_1, y_2, y_3, \cdots, y_n)$$

We can say that there is a Post Correspondence Solution, if for some $i_1, i_2, \cdots, i_k$, where $1 <= i_j <= n$, the condition $x_{i1} \cdots x_{ik} = y_{i1} \cdots y_{ik}$ satisfies.

Example 1:

Find whether the lists

M = (abb, aa, aaa) and N = (bba, aaa, aa)

have a Post Correspondence Solution?

⇒

	x_1	x_2	x_3
M	Abb	aa	aaa
N	Bba	aaa	aa

Here,

$x_2 x_1 x_3$ = 'aaabbaaa'

and $y_2 y_1 y_3$ = 'aaabbaaa'

We can see that,

$x_2 x_1 x_3 = y_2 y_1 y_3$

Hence, the solution is $i = 2$, $j = 1$ and $k = 3$.

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Example 2:
Find whether the lists M = (ab, bab, bbaaa) and N = (a, ba, bab) have Post Correspondence Solution?

⇒

Solution:

	x_1	x_2	x_3
M	ab	bab	bbaaa
N	a	ba	bab

In this case, there is no solution because,

$|x_2 x_1 x_3| \neq |y_2 y_1 y_3|$ [length is different]

Hence, it can be said that this Post Correspondence Problem is undecidable.

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